

AL MAKTOUM INTL, United Arab Emirates

WMO: 412000

Lat: 24.897N

Lon: 55.161E

Elev: 19

StdP: 101.10

Time Zone: 4.00 (E04)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 11.1	(c) 12.2	(d) -1.1	(e) 3.5	(f) 30.6	(g) 1.0	(h) 4.1	(i) 28.6	(j) 10.9	(k) 20.8	(l) 10.0	(m) 21.9	(n) 3.0	(o) 110	(p) 0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	45.1	22.8	44.0	22.8	42.9	23.1	29.9	35.1	29.2	34.5	28.6	34.3	4.5	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.9	25.5	32.9	28.0	24.2	32.9	27.1	22.9	32.7	99.6	34.8	96.3	34.4	93.2	34.5	32.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.3	(b) 9.2	(c) 8.3	(e) 8.3	(f) 47.6	(g) 1.2	(h) 0.6	(i) 7.4	(j) 48.0	(k) 6.7	(l) 48.4	(m) 6.0	(n) 48.7	(o) 5.1	(p) 49.2		
			DB													
			WB													
				8.3	47.6	1.2	0.6	7.4	48.0	6.7	48.4	6.0	48.7	5.1	49.2	
				5.6	31.1	1.2	0.6	4.7	31.5	4.0	31.9	3.3	32.2	2.4	32.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.4	19.1	20.2	23.3	28.0	32.3	34.5	36.9	36.6	34.0	30.1	24.6	20.6	
	DBStd	6.68	1.85	2.87	3.17	2.90	2.27	1.97	1.73	1.49	1.57	1.68	2.17	1.91	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	27	13	9	2	0	0	0	0	0	0	0	0	3	
	CDD10.0	6717	282	287	413	541	690	734	833	825	719	623	439	330	
	CDD18.3	3701	36	63	157	291	432	484	574	567	469	365	189	74	
	CDH23.3	52659	260	482	1413	3422	6348	7596	9574	9376	7183	4756	1756	494	
	CDH26.7	34303	38	158	617	1871	4093	5231	7094	6898	4834	2696	694	79	
Wind	WSAvg	4.0	3.7	4.2	4.4	4.3	4.5	4.4	4.3	4.2	3.8	3.6	3.6	3.4	
Precipitation	PrecAvg	86	15	15	21	8	2	1	4	4	2	2	3	10	
	PrecMax	203	69	90	92	48	6	9	56	8	16	14	28	58	
	PrecMin	20	0	0	0	0	0	0	0	1	0	0	0	0	
	PrecStd	49	18	20	26	11	1	2	10	2	3	3	6	17	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.2	33.8	37.0	40.2	43.9	45.2	46.9	46.1	44.2	39.9	35.1	30.1	
		MCWB	17.3	17.3	19.1	19.6	20.6	22.3	22.8	22.4	22.0	20.9	19.2	18.0	
	2%	DB	27.9	30.8	34.5	38.8	42.1	43.9	45.2	45.0	43.0	38.2	33.2	28.8	
		MCWB	16.9	17.2	17.9	19.1	21.3	22.4	23.0	22.9	22.0	21.2	19.8	18.3	
	5%	DB	26.1	28.4	32.2	37.0	40.9	42.1	44.0	43.8	41.2	37.1	32.0	27.2	
		MCWB	16.4	16.7	17.5	19.0	20.7	22.9	23.5	23.0	22.4	21.7	19.8	18.5	
	10%	DB	24.7	26.2	30.1	35.2	39.1	40.8	42.5	42.1	40.0	35.9	30.8	26.1	
		MCWB	16.7	16.6	17.6	19.1	21.1	23.4	24.1	23.9	23.3	22.2	20.0	17.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.5	20.3	22.2	24.4	27.2	29.6	30.7	30.9	29.7	28.4	24.0	22.2
			MCDB	24.4	25.3	27.0	29.5	32.3	33.7	36.0	35.5	35.0	31.7	29.5	26.1
2%		WB	19.3	19.5	21.1	23.3	26.3	28.7	29.9	30.0	28.8	27.2	23.1	21.2	
		MCDB	23.9	24.6	26.4	28.7	31.7	33.6	35.3	35.1	33.8	31.7	28.5	25.1	
5%		WB	18.4	18.8	20.4	22.3	25.4	27.9	29.2	29.3	28.1	26.5	22.3	20.5	
		MCDB	23.5	23.9	25.9	28.6	31.7	33.7	34.7	34.8	33.3	31.4	28.1	24.6	
10%		WB	17.6	18.1	19.6	21.3	24.4	27.1	28.5	28.5	27.5	25.8	21.5	19.5	
		MCDB	22.6	23.5	25.3	28.7	31.8	33.5	34.7	34.7	32.9	30.9	27.4	24.0	
Mean Daily Temperature Range		5% DB	MDBR	11.2	11.4	12.3	13.3	14.3	14.1	12.9	13.2	13.9	13.0	12.0	11.6
			MCDBR	13.8	14.2	15.3	16.0	16.2	15.6	14.4	14.7	15.6	14.2	13.5	12.8
	MCWBR		6.1	5.9	5.9	5.6	6.3	6.6	6.5	6.6	6.9	6.3	5.2	6.0	
	5% WB	MCDBR	11.5	11.3	12.0	13.2	14.5	14.1	12.5	12.3	13.3	12.8	11.9	10.9	
		MCWBR	6.2	5.7	5.7	5.8	6.2	6.1	5.9	6.3	6.2	5.7	5.2	5.5	
Clear-Sky Solar Irradiance	taub	0.445	0.503	0.566	0.598	0.589	0.639	0.730	0.698	0.601	0.511	0.456	0.437		
	taud	2.040	1.880	1.734	1.695	1.708	1.649	1.533	1.589	1.733	1.905	2.046	2.070		
	Ebn at Noon	792	773	747	735	739	699	637	656	710	755	775	781		
	Edh at Noon	151	190	231	245	242	254	285	269	228	182	149	141		
All-Sky Solar Radiation	RadAvg	4.15	4.94	5.83	6.66	7.44	7.57	7.08	6.81	6.34	5.54	4.48	3.93		
	RadStd	0.25	0.38	0.39	0.40	0.18	0.15	0.22	0.15	0.11	0.09	0.12	0.24		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	+0.47	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	+150	+149

Nomenclature: See separate page